

Statistics and ANOVA Notes

1 Populations, Samples, Distributions, and Descriptive Statistics

1.1 Population vs Sample

- **Population:** The full set of units under study (e.g., all freshmen at Rutgers, all single-family homes in Middlesex County). [file:1]
- **Sample:** Subset of units actually observed; used to infer about the population. [file:1]
- **Simple random sample (SRS):** Every unit has equal chance of selection; avoids selection bias. Haphazard samples are not SRS. [file:1]

1.2 Descriptive Statistics and Distribution Shape

For quantitative variables, describe: [file:1]

- **Center:** mean, median (location of typical values).
- **Spread:** variance, standard deviation (SD), range, interquartile range (IQR), percentiles.
- **Shape:** symmetric vs right-skewed (long right tail) vs left-skewed (long left tail).
- **Outliers:** values that clearly do not fit the overall pattern.

Bell-shaped (approximately normal) distributions concentrate most data near the mean with relatively thin tails. [file:1]

1.3 Normal Distribution and Standardization

If $X \sim N(\mu, \sigma^2)$, then

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

is a standard normal random variable. [file:1]

The empirical 95% rule: roughly 95% of observations lie in $[\mu - 2\sigma, \mu + 2\sigma]$ for a normal distribution. [file:1]

Standardization via z -scores allows use of standard normal tables or software for probabilities. [file:1]

1.4 Bivariate and Multivariate Data; Correlation

- **Bivariate data:** two quantitative variables per unit (e.g., texting frequency vs driving performance). [file:1]
- **Scatterplot:** plots pairs (x, y) ; visually assesses relationship (direction, form, strength, outliers). [file:1]
- **Correlation r :** numerical measure of strength and direction of linear association; population parameter is ρ . [file:1]

Positive r indicates that when one variable is high, the other tends to be high; negative r indicates the opposite. [file:1] Magnitude near 1 (or -1) indicates strong linear association; near 0 indicates weak or no linear association. [file:1]

2 Sampling Distributions, Standard Error, and Central Limit Theorem

2.1 Sampling Distributions

A **sampling distribution** is the distribution of a sample statistic (e.g., $\bar{X}$) when we repeat sampling many times from the same population with the same sample size. [file:1]

Sample statistics vary across samples (*sampling variability*); this is the basis for standard error and for statistical inference. [file:1]

If samples are random (no bias), the sampling distribution is centered at the true population parameter. [file:1]

2.2 Standard Error (SE)

The **standard error** (SE) of a statistic is the standard deviation of its sampling distribution. [file:1] For a sample mean with population SD σ and sample size n ,

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

In practice, σ is typically unknown and is estimated with the sample SD s . [file:1]

2.3 Central Limit Theorem (CLT)

The CLT states that for many statistics, especially sample means, if n is large enough, the sampling distribution is approximately normal and centered at the population parameter, regardless of the shape of the population distribution. [file:1]

As n grows, the sampling distribution becomes more symmetric and bell-shaped, justifying normal-based methods even for non-normal populations. [file:1]

3 Confidence Intervals for Means (One- and Two-Sample)

3.1 One-Sample Confidence Interval for a Population Mean μ

A general confidence interval (CI) has the form

$$\hat{\theta} \pm (\text{critical value}) \times SE.$$

For a mean: [file:1]

- Large sample or known σ :

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}.$$

- Small sample, σ unknown, normal population:

$$\bar{x} \pm t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}},$$

where $t_{\alpha/2, n-1}$ is from the Student t distribution. [file:1]

Margin of error (ME) for a mean: typically $1.96 s/\sqrt{n}$ for a 95% CI and $2.58 s/\sqrt{n}$ for a 99% CI. [file:1]

3.2 Two-Sample CI for Difference in Means $\mu_1 - \mu_2$

Let $X_1, \dots, X_{n_1}$ and $Y_1, \dots, Y_{n_2}$ be two independent samples from populations 1 and 2, respectively. [file:1] The estimator of $\mu_1 - \mu_2$ is $\bar{x}_1 - \bar{x}_2$.

Equal Variances Case (Pooled t -Interval)

Assume $\sigma_1^2 \approx \sigma_2^2$. Rule of thumb: sample SD ratio in $[0.5, 2]$. [file:1]

Pooled variance:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}, \quad s_p = \sqrt{s_p^2}.$$

Standard error for $\bar{x}_1 - \bar{x}_2$:

$$SE = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}.$$

CI:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, n_1+n_2-2} \cdot s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}. [file:1]$$

Unequal Variances Case (Welch / Non-Pooled)

When variances differ, do not pool; instead, use

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}.$$

Degrees of freedom are approximated by the Satterthwaite formula (a function of s_1^2/n_1 and s_2^2/n_2). [file:1] CI:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, \text{df}_{\text{Welch}}} \cdot SE.$$

3.3 Paired Differences CI

When measurements are paired (e.g., before/after on the same subject), form differences

$$d_i = x_{1i} - x_{2i}, \quad i = 1, \dots, n.$$

Treat d_i as a one-sample problem for the mean difference μ_d . [file:1] The CI is

$$\bar{d} \pm t_{\alpha/2, n-1} \cdot \frac{s_d}{\sqrt{n}},$$

where s_d is the SD of the differences and n is the number of pairs.

4 Hypothesis Testing Fundamentals

4.1 Components of a Test

- **Null hypothesis H_0 :** a statement of no effect or no difference (e.g., $\mu = \mu_0$, $\mu_1 = \mu_2$). [file:1]
- **Alternative hypothesis H_a :** the statement that contradicts H_0 , representing an effect or difference. [file:1]
- **Test statistic:** a function of data (e.g., z , t , F) measuring discrepancy between data and H_0 . [file:1]
- **Significance level α :** long-run probability of Type I error (commonly 0.05). [file:1]

4.2 Errors and p-Values

- **Type I error:** rejecting H_0 when it is true; probability α . [file:1]
- **Type II error:** failing to reject H_0 when H_a is true. [file:1]

The **p-value** is the probability, assuming H_0 is true, of observing a test statistic at least as extreme (in the direction of H_a) as the one actually observed. [file:1]

Decision rule:

- If p-value $\leq \alpha$: reject H_0 .
- If p-value $> \alpha$: do not reject H_0 . [file:1]

4.3 One-Sided vs Two-Sided Tests

- One-sided test (e.g., $H_a : \mu > \mu_0$ or $\mu < \mu_0$): p-value is the tail area beyond the observed test statistic in the specified direction. [file:1]
- Two-sided test ($H_a : \mu \neq \mu_0$): p-value is twice the one-sided tail area based on the sign of the statistic. [file:1]

5 One-Sample and Two-Sample t-Tests (Means)

5.1 One-Sample t-Test

Use when population SD is unknown and the population is normal or n is large. [file:1] The test statistic is

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad \text{df} = n - 1.$$

5.2 Two-Sample t-Tests (Independent Samples)

- **Pooled t-test:** assumes equal variances; uses pooled variance s_p^2 and df $n_1 + n_2 - 2$. [file:1]
- **Welch t-test:** assumes unequal variances; uses separate variances and the Satterthwaite df. [file:1]

Check independence of groups, approximate normality within groups, and variance equality rule of thumb before choosing the procedure. [file:1]

5.3 Paired t-Test

For paired data, test hypotheses about mean difference μ_d . The test statistic is the one-sample t on the differences:

$$t = \frac{\bar{d} - d_0}{s_d/\sqrt{n}}, \quad \text{df} = n - 1,$$

where d_0 is typically 0. [file:1]

6 Normality and Equal Variance Diagnostics

6.1 Normality Diagnostics

- **Histograms** of residuals or raw data: check for symmetry and outliers. [file:1][file:3]
- **Normal Q-Q plots:** plot sample quantiles vs theoretical normal quantiles; near-linear patterns suggest approximate normality. [file:1][file:3]

- Theoretical rationale: ordered standard normal variables have expected values at certain quantiles; plotting observed values against these approximate expectations yields a straight line if the data are normal. [file:1]

Formal tests: [file:1][file:3][file:4]

- Shapiro–Wilk: high power, good for small samples.
- Anderson–Darling: emphasizes tails, requires extra package.
- Kolmogorov–Smirnov: less powerful, assumes known μ and σ .
- Ryan–Joiner: similar to Shapiro–Wilk.

t- and F-tests are somewhat robust to mild non-normality, especially in balanced designs. [file:3][file:4]

6.2 Equal Variance (Homoscedasticity) Diagnostics

- Rule of thumb: if one SD is more than twice another SD, equal variance is questionable. [file:2]
- Two-group case: F-test uses the ratio of sample variances; it is sensitive to non-normality. [file:1]
- Multiple groups:
 - Bartlett’s test: powerful if normal, but sensitive to non-normality. [file:2]
 - Levene’s test: uses absolute deviations from group centers; more robust to non-normality but slightly less powerful under perfect normality. [file:1][file:2]

Residual plots and Q–Q plots are used to visually assess homoscedasticity and normality; transformations can help with severe variance heterogeneity. [file:4]

7 One-Way ANOVA (Fixed Effects, Completely Randomized Design)

7.1 Problem Setup

We compare means across $k > 2$ groups (treatments/levels of a factor). [file:2][file:4] Examples include mean cuckoo egg length across nest species or mean dog food sales at different shelf heights. [file:2][file:4]

Fixed-effects single-factor model:

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i = 1, \dots, a, \quad j = 1, \dots, n_i,$$

where μ is the overall mean, τ_i is the treatment effect with $\sum_{i=1}^a \tau_i = 0$, and $\varepsilon_{ij} \sim N(0, \sigma^2)$ independently. [file:4]

7.2 Hypotheses and Sums of Squares

Hypotheses: [file:2]

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k, \quad H_a : \text{at least one mean differs.}$$

Sums of squares: [file:2][file:4]

$$SST = \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{..})^2,$$

$$SSG = \sum_{i=1}^k n_i (\bar{y}_{i.} - \bar{y}_{..})^2,$$

$$SSE = \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i.})^2,$$

with $SST = SSG + SSE$.

7.3 ANOVA Table and F-Test

The ANOVA table has the form: [file:2][file:4]

Source	df	Sum Sq	Mean Sq	F
Treatment	$k - 1$	SSG	$MSG = SSG/(k-1)$	$F = MSG/MSE$
Error	$n - k$	SSE	$MSE = SSE/(n-k)$	—
Total	$n - 1$	SST	—	—

Under H_0 and assumptions, F follows an F distribution with $(k - 1, n - k)$ degrees of freedom. [file:2] Large F indicates that between-group variability is large relative to within-group variability, providing evidence against H_0 . [file:2]

7.4 Assumptions for One-Way ANOVA

- Independent observations. [file:2]
- Normality of errors within each group (checked via Q–Q plots, Shapiro–Wilk). [file:2][file:4]
- Equal variances across groups (checked via SD ratios, Bartlett’s test, Levene’s test). [file:2]

Balanced designs (equal group sizes) improve robustness of ANOVA to assumption violations. [file:3][file:4]

7.5 ANOVA and Regression Equivalence

A single-factor ANOVA with a levels is equivalent to regression with $a - 1$ indicator (dummy) variables. [file:4] The regression model

$$Z = \beta_0 + \beta_1 x_1 + \cdots + \beta_{a-1} x_{a-1} + e$$

leads to the same F-statistic and MSE as the ANOVA model, with each β_i representing a contrast relative to a baseline level. [file:4]

8 Confidence Intervals and Multiple Comparisons in ANOVA

8.1 CIs for Treatment Means and Differences

For treatment i with n_i observations, the CI for μ_i is [file:4]

$$\bar{y}_{i.} \pm t_{\alpha/2, N-a} \sqrt{\frac{\text{MSE}}{n_i}},$$

where N is total sample size and a is number of treatments.

For $\mu_i - \mu_j$, the CI is [file:4]

$$(\bar{y}_{i.} - \bar{y}_{j.}) \pm t_{\alpha/2, N-a} \sqrt{\text{MSE} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}.$$

8.2 Familywise Type I Error and Bonferroni

If you run K tests each at level α , the familywise Type I error (probability of at least one false rejection) is

$$1 - (1 - \alpha)^K \approx K\alpha$$

for small α . [file:4]

For all pairwise comparisons of a means, $K = \binom{a}{2}$; the familywise error can be substantial (e.g., about 0.26 when $a = 4$, $\alpha = 0.05$). [file:4]

The Bonferroni correction uses $\alpha^* = \alpha/K$ for each individual test to guarantee familywise error at most α . [file:4] Equivalently, each CI is constructed with confidence level $1 - \alpha/K$.

8.3 Post-hoc Procedures

- **Bonferroni pairwise t-tests:** simple but conservative; implemented via functions that apply Bonferroni-adjusted CIs or p-values. [file:4]
- **Holm's method:** more powerful step-down procedure; often default in pairwise t-test implementations. [file:4]

- **Tukey’s HSD:** uses the studentized range distribution; controls familywise error for all pairwise comparisons efficiently. [file:4]
- **Fisher’s LSD:** uses simple t-tests after a significant overall ANOVA; offers less control over familywise error and is typically recommended only with a small number of comparisons. [file:4]

9 Diagnostics, Transformations, and Sample Size in ANOVA

9.1 Residual Diagnostics

- **Residual vs fitted plot:** residuals should scatter randomly about 0 with roughly constant spread; funnel shapes suggest heteroscedasticity. [file:4]
- **Normal Q–Q plot of residuals:** checks for normality; major deviations in tails indicate non-normality. [file:4]

Studentized residuals help identify outliers; a few moderate outliers are often tolerable, but severe ones may require special handling. [file:4]

9.2 Transformations for Heteroscedasticity

When variance grows with mean (e.g., counts, Poisson-like), transformations such as square root or log can stabilize variance. [file:4]

Example: counts of flaws under different coatings show unequal variances and a funnel pattern in residuals; a square root transformation of counts yields better- behaved residuals and more appropriate ANOVA. [file:4]

9.3 Sample Size Planning for ANOVA

To plan sample size, specify: [file:4]

- Number of groups a .
- Within-group variance (MSE) and expected effect size (between-group variability).
- Type I error α .
- Desired power $1 - \beta$.

Analytical formulas or software (e.g., power calculations) can determine required per-group sample size based on these inputs. [file:4]

10 Blocking and Randomized Complete Block Design (RCBD)

10.1 Motivation for Blocking

Many experiments have nuisance variables that cause variability but are not of primary interest. [file:3] These factors may be:

- Known and controllable: handled via blocking (e.g., location, manufacturer, gender). [file:3]
- Known but uncontrollable: handled using covariates in ANCOVA/regression. [file:3]
- Unknown or uncontrollable: mitigated by randomization. [file:3]

Blocking groups experimental units into homogeneous sets to reduce within-block variability and isolate nuisance variation. [file:3]

10.2 RCBD Structure and Model

In an RCBD:

- Each block contains all treatments (complete replicate).
- Within each block, treatment assignments are randomized.
- Blocks represent levels of a nuisance factor. [file:3]

Fixed-effects RCBD model:

$$Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij},$$

where τ_i is the treatment effect, β_j is the block effect, and $\varepsilon_{ij} \sim N(0, \sigma^2)$. [file:3]

RCBD can be viewed as an extension of the paired t-test to more than two treatments. [file:3]

10.3 RCBD ANOVA and Interpretation

RCBD ANOVA table has sources: Treatment, Block, Error, and Total. [file:3] The F-test for treatments examines mean differences after adjusting for block effects. The F-test for blocks assesses whether block means differ, indicating the presence of block-related variability. [file:3]

Example: in an apple orchard experiment, blocking by location reduces error variance compared to a completely randomized design, even if treatment differences are not significant. [file:3]

10.4 Model Adequacy with Blocking

Residual diagnostics in RCBD also examine residuals versus blocks as well as versus fitted values. [file:3] Systematic patterns across blocks may indicate treatment-block interactions or violations of additive assumptions.

Normality and homoscedasticity checks remain the same as in CRD ANOVA. [file:3]

11 Random Effects One-Way ANOVA and Intraclass Correlation

11.1 When the Factor is Random

In a random-effects model, the levels of the factor are considered a random sample from a larger population of possible levels. [file:4] Inference focuses on variance components rather than on specific level means.

Example: sire in an animal inheritance study is treated as a random effect when sires represent a sample from a population of sires. [file:4]

11.2 Random Effects Model and Variance Components

Random-effects one-way ANOVA model:

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij},$$

with $\alpha_i \sim N(0, \sigma_\alpha^2)$, and $\varepsilon_{ij} \sim N(0, \sigma^2)$, independent. [file:4]

Total variance decomposes as:

$$\text{Var}(Y_{ij}) = \sigma_\alpha^2 + \sigma^2.$$

Expected mean squares: [file:4]

$$E(MSTreatment) = \sigma^2 + n\sigma_\alpha^2,$$

$$E(MSE) = \sigma^2.$$

Estimators:

$$\hat{\sigma}^2 = MSE, \quad \hat{\sigma}_\alpha^2 = \frac{MSTreatment - MSE}{n}.$$

Negative estimates of σ_α^2 can occur if the true treatment variance is very small or sample size is small. [file:4]

11.3 Intraclass Correlation Coefficient (ICC)

The intraclass correlation coefficient is

$$\rho = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma^2},$$

representing correlation between two observations from the same level of the random factor. [file:4]

Interpretation: [file:4]

- $\rho = 0$: no group structure (all variation is within groups).
- $\rho \rightarrow 1$: observations within a group are almost identical compared to differences between groups.

CIs for σ^2 and ρ can be constructed using F-distribution-based methods; CIs for σ_α^2 itself are more complex. [file:4]

12 Nonparametric Alternatives

12.1 Wilcoxon Signed Rank Test

The Wilcoxon signed rank test compares a population median (or paired differences) to a specified value without assuming normality, but assumes symmetry. [file:4]

Procedure: [file:4]

1. Compute differences $d_i = X_i - m_0$ (or pairwise differences).
2. Discard zero differences, rank absolute differences from smallest to largest.
3. Assign signs to ranks, form the sum of positive ranks W .
4. Under H_0 , W has a known distribution; for moderate n , approximate by a normal distribution with

$$E(W) = \frac{n(n+1)}{4}, \quad \text{Var}(W) = \frac{n(n+1)(2n+1)}{24}.$$

Small W in a lower-tailed test suggests evidence that the median is less than m_0 . [file:4]

12.2 Mann–Whitney (Wilcoxon Rank Sum) Test

The Mann–Whitney test compares two independent groups when distributions have similar shape but may not be normal. [file:4]

Procedure: [file:4]

1. Combine both samples, rank all observations.
2. Compute the sum of ranks for one group, denoted W_1 .
3. Transform to U statistic; for group sizes m and n ,

$$E(U_1) = \frac{mn}{2}, \quad \text{Var}(U_1) = \frac{mn(m+n+1)}{12}.$$

4. Approximate U_1 as normal and compute a z-score for p-value.

The test evaluates whether one population tends to yield larger observations than the other (location shift). [file:4]

12.3 Kruskal–Wallis H Test (Nonparametric ANOVA)

The Kruskal–Wallis test is a nonparametric alternative to one-way ANOVA for comparing k groups. [file:4]

Procedure: [file:4]

1. Combine all observations, rank them (ties get average ranks).
2. Let T_i be the sum of ranks in group i , n_i be group sizes, and N total sample size.
3. Compute

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{T_i^2}{n_i} - 3(N+1).$$

Under H_0 that all distributions are identical, H is approximately χ_{k-1}^2 . Large H indicates at least one group's distribution differs. [file:4]